Na Min An

Ph.D. Student

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Research Interests

My research aims to explore: *how do omni-modal AI systems perceive, integrate, and reason over the full spectrum of human sensory experience?* I am deeply interested in uncovering the mechanisms behind multisensory processing, such as complex textual, audio, and scene understanding and reasoning.

Education

Korea Advanced Institute of Science and Technology (KAIST) *Seoul, South Korea*
Doctor of Philosophy in Artificial Intelligence, GPA: 4.13/4.3 *Feb.2023-Current*
- Advisors: Prof. Hyunjung Shim (2024-Current) and Dr. James Thorne (2023-2025)

University of Seoul *Seoul, South Korea*
Bachelor of Science in Mathematics, GPA: 4.29/4.5 *Feb. 2022*
- Summa Cum Laude (1/30) and Received academic excellence or Mayor's scholarship for every semester)

Skills

Language & Framework Python (PyTorch, HuggingFace Transformers), C++, JavaScript, MATLAB, R; Linux, Cloud/Server Computing
NLP & Multimodal LLM LLM/VLM training, Image Captioning, Text-to-Image Generation, Video Understanding, Multimodal Reasoning
Computer Vision Scene Understanding, Language-Guided Visual Grounding, Multimodal Retrieval, Data Curation at Scale
Interpretability & Fairness Mechanistic Interpretability, Human Alignment Metrics
Certifications Society of Actuaries (P, FM, and SRM)

Experience

KAIST *Seoul, South Korea*
Research Assistant *March 2023 - PRESENT*
- Ministry of Science and ICT Project: Contributed leadership to a national project aimed at developing socially unbiased generative models.
- Basic Research Lab Project: Co-led a national project on developing barrier-free AI systems in educational and mobility domains.
- Published eight first-author and three co-authored Top-tier works, with research centered on fairness and human-centered ML systems.

NAVER AI Lab *Seongnam, South Korea*
Research Intern *April 2026 - PRESENT*
- Analyzing the architecture of omni-modal models and modality effects

University of Copenhagen *Copenhagen, Denmark*
Visiting Researcher *March 2025 - May 2025*
- Received a Danish Data Science Academy grant to conduct research on interpretability in multimodal models towards fairness-aware systems.
- Developed a neuron-level bias mitigation framework for VLMs and LVLMs, in collaboration with the NVIDIA Taiwan team.

Korea Institute of Science and Technology (KIST) *Seoul, South Korea*
Research Intern *Oct. 2022 - Feb. 2023*
- Analyzed human perceptual limits and ML model performance under severely low-resolution visual conditions.
- Developed a reinforcement learning–based framework that emulates the learning effects frequently occurring in human psychophysical tests.
- Built an automated imaging pipeline for detecting fluorescence-image edges and identifying hyperpolarization phenomena with high precision.

University of Seoul *Seoul, South Korea*
Research Intern *Sep. 2020 - Dec. 2020*
- Investigated district-level Covid-19 risk factors across Seoul and awarded 3rd place (out of 40+ teams) in a competitive big-data competition.
- Developed a hybrid CNN–RNN model to classify the progressive stages of pulmonary nodules from sequential CT scan slices.
- Analyzed YouTube channel performance to model subscriber trajectory and provided targeted strategies for subscriber acquisition.

Selected Publications

1. Kim, E.*, **An, N. M.***, Kang, W. J., Kim, S., Thorne, J., and, Shim, H., "Are Large Vision-Language Models Ready to Guide Blind and Low-Vision Individuals?," *COLM* (* denotes equal contribution), 2026.
   Accessibility | Vision-Language Models | Evaluation & Benchmarking
2. **An, N. M.***, Kang, I.*, Lee, M.„ and Shim, H., "Blind to Position, Biased in Language: Probing Mid-Layer Representational Bias in Vision Language Encoders for Zero-Shot Language-Grounded Spatial Understanding," *ECCV* (* denotes equal contribution), 2026.
   Vision-Language Models | Interpretability
3. **An, N. M.**, Jang Y., Hirota, Y., Hachiuma, R., Augenstein, I., and Shim, H., "Interpretable Debiasing of Vision-Language Models for Social Fairness," *CVPR*, 2026.
   Interpretability | Fairness & Bias | Vision-Language Models | Evaluation & Benchmarking
4. Kim, E.*, Park, J.*, **An, N. M.***., Kim, J., Patel, H. L., Jin, J., Kruk, J., Agarwal, A., Panda, S., Ilasariya, F. A., Shim, H., and Oh, A., "World in a Frame: Understanding Culture Mixing as a New Challenge for Vision-Language Models," *CVPR* (* denotes equal contribution), 2026.
   Fairness & Bias | Vision-Language Models | Evaluation & Benchmarking
5. Waheed, S.* and **An, N. M.***, "Image Embedding Sampling Method for Diverse Captioning," *EMNLP Main* (* denotes equal contribution), 2025.
   Multimodal Representation | Vision-Language Models
6. **An, N. M.***, Kim, E.*, Thorne, J., and Shim, H., "I0T: Embedding Standardization Method Towards Zero Modality Gap," *ACL Outstanding* (Top 1%, 26 out of 3,000 accepted papers, * denotes equal contribution), 2025.
   Multimodal Representation | Vision-Language Models | Evaluation & Benchmarking
7. Bayramli, Z.*, Suleymanzade, A.*, **An, N. M.**, Ahmad, H., Kim, E., Kang, J., Thorne, J., and Oh, A., "Diffusion Models Through a Global Lens: Are They Culturally Inclusive?" *ACL Oral* (Top 8%, 243 out of 3,000 accepted papers, * denotes equal contribution), 2025.
   Fairness & Bias | Evaluation & Benchmarking
8. Kang, W. J., Kim, E., **An, N. M.**, Kim, S., Choi, H., Kwak, K., and Thorne, J., "Sightation Counts: Leveraging Sighted User Feedback in Building a BLV-aligned Dataset of Diagram Descriptions," *ACL Main*, 2025.
   Accessibility | Evaluation & Benchmarking
9. **An, N. M.**, Roh, H., Kim, S., Kim, J. H., and Im, M., "Machine Learning Techniques for Simulating Human Psychophysical Testing of Low-Resolution Phosphene Face Images in Artificial Vision," *Advanced Science* (Impact Factor: 14.3, JCR Ranking: 6.5%), 2025 - Featured in *Korean National News*.
   Accessibility | Evaluation & Benchmarking
10. Jung, W., Hong, J., **An, N. M.**, Thorne, J, and Yun, S., "Stable Language Model Pre-training by Reducing Embedding Variability," *EMNLP Main*, 2024.
    Language Models
11. Lee, N.*, **An, N. M.***, and Thorne, J. "Can Large Language Models Infer and Disagree Like Humans?" *EMNLP Main* (* denotes equal contribution), 2023.
    Language Models | Evaluation & Benchmarking

Publications

1. Kim, E.*, **An, N. M.***, Kang, W. J., Kim, S., Thorne, J., and, Shim, H., "Are Large Vision-Language Models Ready to Guide Blind and Low-Vision Individuals?," *arXiv:2510.00766* (* denotes equal contribution), 2026.
2. Kim, E.*, Park, J.*, Park, N*, **An, N. M.***, Kim, K.*, Yoon, S., Huo, J., and, Shim, H., "Aligned but Stereotypical? The Hidden Influence of System Prompts on Social Bias in LVLM-Based Text-to-Image Models," *ICLR RSI Workshop* (* denotes equal contribution), 2026.
3. Waheed, S., **An, N. M.**, Milford, M., Ramchurn, S. D., and Ehsan, S., "VLM-Guided Visual Place Recognition for Planet-Scale Geo-Localization," *ACRA*, 2025.
4. Kim, J. S*, Yu, A.*, Ismayilzada J.*, Park, J., Kim, E., Ahmad, H., **An, N. M.**, Thorne, J., and Oh, A., "WHEN TOM EATS KIMCHI: Evaluating Cultural Awareness of Multimodal Large Language Models in Cultural Mixture Contexts," *NAACL C3NLP Workshop Outstanding* (* denotes equal contribution), 2025.
5. **An, N. M.**, Weed, S., and Thorne, J., "Capturing the Relationship Between Sentence Triplets for LLM and Human-Generated Texts to Enhance Sentence Embeddings," *EACL Findings*, 2024.
6. **An, N. M.**, Roh, H., Kim, S., Kim, J. H., and Im, M. "Reinforcement Learning Framework to Simulate Short-Term Learning Effects of Human Psychophysical Experiments Assessing the Quality of Artificial Vision," *IJCNN Oral*, 2023.
7. **An, N. M.**, Roh. H., Jung, S., Kim, E. J., and Im, M. "Machine Learning Can Approaches as an Alternative to Human Psychophysical Tests of Prosthetic Vision," *EMBS Abstract*, 2021.

Patents

1. Im, M., Roh, H., and **An, N. M.**, Kim, J. H., "Artificial Vision Parameter Learning and Automating Method for Improving Visual Prosthetic Systems", *US Patent Granted No. US 12,602,618 B2*, 2026.
2. Im, M., Roh, H., and **An, N. M.**, Kim, J. H., "Artificial Vision Expression Parameter Automation Learning System Method for Improving Artificial Vision Device", *Korea Patent Granted No. 0172619*, 2021.
3. **An, N. M.**, Kim, E., Shim, H., "A Fine-Grained Vision-Language Alignment Evaluation Framework Reflecting Users' Diverse Visual Abilities", *KAIST Patent Submitted No. P2026-0725*, 2026.

Service

Talks

- AI Tutorial Series at KAIST AI (Winter 2025).

Volunteer

- Provided on-site support for EMNLP conference (2023).
- Contributed to community support and educational outreach in Cambodia in Summer 2018.

Team Leader

- Lab representative student from March 2024 to Aug. 2025.
- Appeared in the KAIST AI promotional video (2024) and top-10 research achievements participant (2023).
- Lab research manager from Aug. 2023 to Dec. 2023.

Teaching Assistant

- Department Seminars & Colloquia, Fall 2023 & Spring 2026
- Computational Image Generation and Modification, Fall 2025

Mentor

- Led research mentorship for two Azerbaijan undergraduate students in South Korea from Sep. 2024 to Jan. 2025.
- Provided English and mathematics tutoring to Dongdaemun-gu (Seoul, South Korea) high school students in Fall 2019 and Winter 2020.

Reviewer

- IEEE Transactions on Audio, Speech and Language Processing 2025 and 2026
- ACL ARR February, May, July, October 2025; January, April, May, July 2026; COLM 2026
- ACL Student Research Workshop 2025 and 2026; ICLR RSI Workshop 2026
- IJCNN 2024, 2025, 2026, and 2027

References

- **Hyunjung Shim**, Associate Professor, KAIST AI | `kateshim@kaist.ac.kr`
- **Dongyoon Han**, Leader, NAVER AI Lab | `dongyoon.han@navercorp.com`
- **Yusuke Hirota**, Research Scientist, NVIDIA | `yusukeh@nvidia.com`
- **Isabelle Augenstein**, Professor, University of Copenhagen | `augenstein@di.ku.dk`
- **Alice Oh**, Professor, KAIST School of Computing | `alice.oh@kaist.edu`
- **James Thorne**, Chief AI Scientist, Theia Insights | `james@jamesthorne.com`
- **Maesoon Im**, Principal Research Scientist, KIST | `maesoon.im@kist.re.kr`